

FINAL EXPLANATION: MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: MATHEMATICS GRADE 10 – SUB-STRAND 3.4 LINEAR MOTION: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for the unit. Driving question: How can mathematics help us fully describe, graph, and predict the motion of objects moving in a straight line, and why does direction always matter? Answer every section in the space under the prompt. Use full sentences, show every step of any calculation, and include units. Refer back to the Timer Race Puzzle, the Driving Question Board and your cumulative motion model from Lessons 1 to 10 where you can. Sketch graphs neatly and label both axes. You may use the formulae $v = s/t$ and $a = (v - u)/t$, and what you learned about graphs.

Part 1: Solving the Timer Race Puzzle

In the Timer Race, Group A walked a straight line, Group B walked a zigzag path and Group C walked the zigzag path and then retraced their steps back to a midpoint. Group C covered a similar total distance to the others, yet ended up in a completely different place. (a) Explain why this happened using the terms distance, displacement, scalar and vector. (b) Eliud Kipchoge runs 42 km in a marathon yet ends only a few hundred metres from where he started. Use your explanation from (a) to say what this tells us about his distance and his displacement. (c) Why is knowing the distance alone not enough to say where a traveller ends up?

Write your answer here:

Part 2: Speed, velocity and the matatu speedometer

(a) A matatu driver says, 'My speedometer shows 60 km/h.' Explain why this reading cannot tell you which direction the matatu is heading. **(b)** Write the formula for speed and the formula for velocity, and say which one is a vector. **(c)** A matatu makes a return trip along Thika Road and ends at its starting point. What are its displacement and its average velocity for the whole trip? Is its average speed also zero? Explain.

Write your answer here:

Part 3: Reading motion from a displacement–time graph

Walker C in the Timer Race walked away from the start, then retraced steps back towards the start. (a) Describe in words what the displacement–time graph for Walker C would look like. Say what a positive gradient and a negative gradient each mean. (b) What does a horizontal section of a displacement–time graph tell you? (c) Explain why this graph shows direction, but a speedometer or a distance value does not.

Write your answer here:

Part 4: Acceleration and the boda-boda stopping distance

A boda-boda is travelling at 30 km/h and brakes uniformly until it stops in 5 seconds. (a) Convert 30 km/h to m/s, giving your answer to 2 decimal places. (b) Calculate the acceleration using $a = (v - u) / t$. Explain what the sign of your answer means. (c) Sketch the velocity–time graph for the braking. Use the area under the graph to find the stopping distance, and give your answer to the nearest metre. (d) Why does the boda-boda have a negative acceleration even though it is still moving forward?

Write your answer here:

Part 5: Relative speed and the final answer to the driving question

Two matatus travel on the Nairobi–Mombasa highway. (a) State the rule for the relative speed of the two matatus when they travel in the same direction, and the rule when they travel in opposite directions. Use v_1 and v_2 for their speeds. (b) Explain why direction decides which rule you use. (c) Using the Timer Race Puzzle, the marathon and at least two kinds of graph or formula from this unit, write a short answer to the driving question: How can mathematics help us fully describe, graph and predict motion in a straight line, and why does direction always matter?

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Distinguishing scalars and vectors (distance/displacement, speed/velocity)	Defines all four terms correctly, labels each as scalar or vector, and uses them accurately to explain the Timer Race Puzzle and Kipchoge's marathon.	Defines most terms correctly and uses them to explain the puzzle, with minor gaps in scalar/vector labelling.	Mixes up distance and displacement or speed and velocity, or does not link them to the puzzle.
Calculations and units	Converts units correctly, applies $a = (v - u)/t$ and the area under the graph correctly, gives the sign and units, and shows every step.	Applies the formulae correctly with small slips in units, rounding or sign, and shows most steps.	Uses the wrong formula, omits units or the sign, or does not show working.
Interpreting graphs	Explains that the gradient of a displacement–time graph is velocity and the gradient of a velocity–time graph is acceleration, interprets negative and zero gradients, and uses area as displacement.	Interprets gradients correctly but is less clear on negative gradients or on area as displacement.	Describes the graph shape only, or confuses displacement–time and velocity–time graphs.
Explaining why direction matters	Consistently links the sign of velocity, acceleration and gradient to direction, and includes relative speed rules with a clear reason for adding or subtracting.	States that direction matters and gives some correct links to signs or relative speed.	Says direction matters without evidence or explanation.

Use of evidence and the driving question	Draws on the Timer Race Puzzle, Kipchoge, the matatu and the boda-boda to give a complete, well-organised answer to the driving question.	Uses some unit examples to answer the driving question with reasonable organisation.	Gives a general answer with few or no examples from the unit.
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ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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